MAITRI: A Multimodal AI Companion for Enhancing Astronaut Psychological and Physical Well-Being in Deep-Space Missions

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***Abstract*—Long-duration deep-space missions subject astronauts to compound physiological and psychological stressors including microgravity-induced deconditioning, circadian disruption, social isolation, and communication latency exceeding 44 minutes at Mars conjunction. This paper presents MAITRI (Mental and Agentic Intelligence for Total Resilience Integration), a modular multimodal AI companion system for autonomous astronaut health support. The physiological stress detection pipeline is trained on the WESAD benchmark dataset and evaluated using subject-independent Leave-One-Subject-Out (LOSO) cross-validation across all 15 subjects, yielding 81.61% accuracy and 81.05% weighted F1 for a class-balanced Random Forest classifier, with a per-subject mean of 81.50% ± 13.68%. This is within the subject-independent generalisation range reported in [10], where the original authors achieved 88.33% (wrist RF) and 92.01% (chest RF) on the binary task using LOSO evaluation. SVM achieves 77.91% LOSO accuracy. Full per-class metrics are reported: calm class precision/recall/F1 = 82%/91%/86%; stress class = 81%/64%/72%. Feature attribution analysis identifies respiratory amplitude (0.1535), EDA standard deviation (0.1060), and mean skin temperature (0.0731) as the three most discriminative stress biomarkers. The text pipeline employs a pre-trained mental health NLP model in zero-shot inference with linguistic features and a lexicon-based fallback. A RAG pipeline using vector retrieval and a locally deployed LLM generates evidence-grounded responses fully on-device. MAITRI achieves sub-2-second response latency and 4.6/5 user satisfaction in an initial usability evaluation. This work is presented as an engineering integration contribution advancing autonomous AI health support for crewed space exploration.**

***Index Terms*—Astronaut health monitoring; multimodal stress detection; WESAD; LOSO cross-validation; Random Forest; MentalBERT; retrieval-augmented generation; SHAP explain- ability; deep-space missions; autonomous AI companion; human- machine systems; explainable AI**

I. Introduction

The space exploration enterprise has entered a transformative era. National space agencies and commercial operators are actively planning crewed missions to cislunar space, the lunar surface, and Mars, with transit durations of six to nine months and surface-stay durations exceeding one year [1]. These missions will place human crews in environments that are profoundly hostile to biological and psychological functioning—not for days or weeks as in prior missions, but for years. The physiological and psychological demands of such missions are qualitatively different from anything in the history of human spaceflight.

The physiological consequences of extended spaceflight are well-characterised and serious. Bone mineral density decreases at approximately 1–2% per month in load-bearing skeletal sites in the absence of countermeasures [2]. Skeletal muscle atrophy occurs progressively throughout the mission, cardiovascular deconditioning reduces aerobic capacity, neurovestibular adaptation creates persistent orientation disturbances, and cephalad fluid shift is associated with visual impairment including papilloedema and optic disc flattening—the VIIP (Visual Impairment and Intracranial Pressure) syndrome documented in a significant proportion of long-duration ISS crew members [3]. Each of these changes represents not merely a health concern but a mission-critical performance risk: a crew member experiencing severe fatigue, musculoskeletal pain, or visual impairment may be unable to perform an extravehicular activity, operate critical equipment, or respond to an emergency with full cognitive and physical capacity.

The psychological burden of deep-space missions is equally significant, and perhaps less well-understood by the general

public. Studies conducted on ISS crews and behavioral analog environments including HI-SEAS (Hawaii Space Exploration Analog and Simulation), MARS-500, Concordia Antarctic station, and nuclear submarine deployments document a characteristic progression of psychological challenges: elevated anxiety and irritability during mission inception as crews adapt to confinement; interpersonal conflict and social friction in mid-mission as the novelty of the environment fades; depressive symptomatology, motivational decline, and cognitive performance decrements in the latter phases of long missions [4], [5]. Teamwork deteriorates precisely when mission complexity is highest, creating a dangerous intersection of human factors and operational demand. For Mars missions lasting 18–30 months, this psychological arc will play out far from any possibility of Earth-based psychological intervention.

A constraint unique to deep-space operations that has no precedent in the ISS era is communication latency. At Mars opposition, one-way signal delay reaches approximately 3 minutes; at conjunction it exceeds 22 minutes, with round-trip delays exceeding 44 minutes [6]. This renders real-time consultation with Earth-based medical personnel, mission psychologists, and ground control impossible. An astronaut experiencing a medical or psychological crisis cannot receive timely guidance. This single operational reality transforms the requirement for onboard health management from a telemetry relay function—as it is today on the ISS—to a fully autonomous diagnostic and support capability. The spacecraft must carry, in software, the health monitoring and psychological support functions that Earth-based teams currently provide in real time.

Current onboard health-monitoring infrastructure, while technically capable of capturing physiological telemetry, remains fundamentally a sensor aggregation and relay system. Data are transmitted to Earth for expert interpretation; recommendations flow back with multi-minute delays. No deployed system provides autonomous multimodal stress detection, evidence-grounded psychological support, explainable AI prediction rationale, or longitudinal behavioral trend analysis in a single on-device architecture [7]. The closest deployed precedent relies on cloud-based NLP that would be non-functional during deep-space communication blackouts [9]. The gap between the health management systems currently deployed on the ISS and those required for Mars is large and not closing fast enough.

This paper presents MAITRI—Mental and Agentic Intelligence for Total Resilience Integration—as a contribution toward closing this gap. MAITRI is a software-based multimodal AI companion system that integrates physiological stress classification, natural language processing, retrieval-augmented generation, and explainable AI into a unified, fully on-device architecture. It is designed to monitor astronaut stress and well-being continuously, provide evidence-grounded psychological support proactively, and do so without any dependency on Earth-based communication or cloud computing infrastructure. The system is presented as an engineering integration contribution: its novelty lies not in the invention of new algorithms but in the purposeful combination of proven components into an architecture designed for the specific operational and safety constraints of deep-space crewed missions.

The principal technical and scientific contributions of this work are:

- A subject-independent LOSO evaluation of physiological stress classification on the WESAD benchmark, yielding 81.61% accuracy and 81.05% weighted F1—evaluated using subject-independent LOSO cross-validation, representing honest generalisation performance on unseen individuals.
- Full per-class precision, recall, and F1 reporting (calm: 82%/91%/86%; stress: 81%/64%/72%), including a threshold sensitivity analysis showing that lowering the decision boundary from 0.5 to 0.35 improves stress recall from 64% to 74% for safety-critical deployment.
- SHAP-based global feature importance analysis identifying the five most discriminative physiological stress biomarkers with physiological interpretation, enabling clinically auditable prediction rationale.
- A fully on-device RAG architecture combining vector retrieval and a locally deployed LLM [15] for evidence-grounded response generation from a curated NASA behavioral health knowledge base, satisfying the communication autonomy requirements of deep-space deployment.
- An integrated full-stack system architecture—FastAPI backend, React.js frontend, SQLite persistence with AES-256 encryption, MLflow experiment tracking—demonstrating end-to-end deployability in sub-2-second response latency with 100% uptime over 72-hour continuous operation.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the methodology. Section IV presents the system architecture. Section V reports results. Section VI discusses findings, limitations, and future work. Sections VII, VII, and VII present the conclusion, acknowledgment, and references.

II. Related Work

A. Astronaut Health Monitoring Systems

Contemporary ISS health monitoring relies on the NASA HRP data ecosystem, aggregating wearable biosensor streams including actigraphy, HRV, skin conductance, and EEG for Earth-side expert analysis [2]. Prior work has demonstrated actigraphy-based sleep deficiency characterisation in ISS crews with associations to cognitive performance decrements [8]. A free-floating AI assistant deployed on the ISS supports procedural guidance through cloud-based NLP [9]. Its cloud-dependent architecture is operationally unsuitable for deep-space communication blackouts. MAITRI's fully on-device pipeline directly addresses this limitation.

B. Physiological Stress Detection on WESAD

The WESAD dataset [10] is the primary physiological benchmark for wearable stress detection. The WESAD benchmark paper [10] reports a subject-independent Random For-

est achieving 88.33% on wrist physiological signals and 92.01% on chest physiological signals (both binary, LOSO evaluation). This constitutes the most directly comparable subject-independent baseline for MAITRI's LOSO evaluation. MAITRI achieves 81.61% LOSO accuracy using 15-feature extraction from high-resolution chest sensors via neurokit2 [16], consistent with the subject-independent generalisation range established in [10]. Subject-dependent evaluations in the literature report higher figures but are not comparable to LOSO results; all comparisons in this paper clearly label evaluation protocols.

C. AI and NLP in Mental Health Support

A CBT-aligned conversational chatbot demonstrated statistically significant PHQ-9 reductions in an RCT comparable to brief human-delivered CBT [11]. Similar findings were replicated with an empathy-driven conversational agent [12]. MentalBERT [13] provides superior domain alignment to VADER [14] for mental health text classification. In MAITRI, MentalBERT [13] is deployed zero-shot; quantitative NLP evaluation on astronaut-domain text is identified as future work. Trust and transparency in human-autonomous system interaction—a critical consideration for safety-critical AI systems including MAITRI—is addressed through SHAP explainability [21].

D. Retrieval-Augmented Generation

RAG was introduced as a framework for grounding LLM generation in retrieved factual context, reducing hallucination and improving faithfulness [15]. MAITRI applies RAG to ground psychological support responses in a NASA HRP knowledge base, ensuring recommendations are traceable to evidence-based clinical guidelines. Automated RAG evaluation using BERTScore and blind expert relevance ratings is identified as a limitation and future work target.

E. Research Gaps Addressed

Four specific gaps motivate this work: (1) no deployed space health system integrates autonomous multimodal stress detection with psychological support; (2) cloud-dependent AI systems [9] cannot operate during deep-space communication blackouts; (3) prior WESAD evaluations often use subject-dependent protocols, overstating generalisation; (4) no published system combines SHAP explainability with on-device RAG support in a single deployable architecture.

III. Methodology

A. Dataset and Preprocessing

WESAD [10] provides 700 Hz chest-worn recordings from 15 healthy subjects. Binary classification retains baseline (calm, label=0) and stress (label=1) conditions. Subject S12 is absent from WESAD due to data collection issues documented by the dataset creators [10]; the 15 available subjects are S2–S11 and S13–S17. Sliding window segmentation (60 s window, 30 s overlap) yields 919 windows: 587 calm and 332 stress (64/36 imbalance). Class imbalance is addressed via `class_weight='balanced'`, which inversely weights classes by frequency during training without requiring synthetic oversampling.

B. Feature Extraction

Fifteen features are extracted per window using neurokit2 [16]. From ECG: RMSSD, SDNN, mean RR, pNN50. From EDA: mean, std, SCR peaks, tonic mean. From respiration: rate and amplitude. From temperature: mean, std, slope. From EMG: mean absolute value and variance. This feature set spans the principal physiological dimensions of the acute stress response.

C. Subject-Independent Evaluation Protocol

All classification experiments use Leave-One-Subject-Out (LOSO) cross-validation: 15 folds, each leaving one subject as the held-out test set. This strictly prevents physiological signature leakage from test subjects into training data, providing a subject-independent generalisation estimate appropriate for deployment on individuals not represented in training. Subject-dependent results from prior literature are not directly comparable and are labelled explicitly throughout. The LOSO threshold sensitivity analysis in Section III-F was applied to the held-out LOSO fold predictions, not to a separately trained full-dataset model, ensuring consistency with the evaluation protocol.

D. Classification Models

Random Forest (200 trees, max_depth=15, class_weight='balanced') is the primary model. SVM with RBF kernel (C=1.0, γ=scale, class_weight='balanced') serves as the comparator. Both are embedded in scikit-learn `Pipeline` objects with within-fold `StandardScaler` normalisation to prevent data leakage through scaling statistics.

E. SHAP Explainability — Limitation Acknowledged

SHAP TreeExplainer [17] is applied to the Random Forest trained on the full 919-sample dataset to produce a global feature importance ranking. It is acknowledged that this model has seen all subjects; the SHAP values are therefore not strictly subject-independent. Aggregating SHAP values across LOSO fold models is identified as more rigorous and is targeted in future work. The current SHAP analysis is presented as indicative of global feature relevance.

F. Stress Class Recall and Safety Threshold Analysis

The LOSO stress class recall of 64% (false-negative rate 36%) is the primary safety-relevant result. A recall-optimised threshold analysis was applied to the held-out LOSO fold predictions: lowering the stress class decision boundary from 0.5 to 0.35 increases stress recall from 64% to 74% at the cost of precision reduction from 81% to 70%. For safety-critical operational deployment, a threshold of 0.35 is recommended to prioritise recall. MAITRI further mitigates false negatives through multimodal fusion (text and acoustic modalities provide complementary detection pathways), personalised baseline anomaly detection, and longitudinal trend monitoring.

G. NLP Pipeline

MentalBERT [13] is deployed zero-shot for three-class sentiment classification (calm, moderate, stressed). SpaCy `en_core_web_sm` extracts eight linguistic stress indicators. VADER provides a fallback [14]. NLP stress classification accuracy on astronaut-domain text has not been quantitatively evaluated—an acknowledged limitation.

H. RAG Pipeline

A 47-chunk knowledge base from NASA HRP guidelines, CBT protocols, and relaxation techniques is embedded using sentence-transformers and indexed in a FAISS store [15]. Top-3 chunks are retrieved per query and concatenated with user message, stress tier, and last five conversation turns as context for a locally deployed LLM [15]. All inference runs on-device. RAG evaluation using BERTScore and blind human expert relevance ratings is identified as required future work.

IV. System Architecture

A. Five-Layer Architecture

MAITRI adopts a five-layer modular architecture illustrated in Fig. 1. Layer 1 (Input) captures text, voice, physiological CSV, and optional facial expression signals. Layer 2 (Feature Extraction) comprises MentalBERT, Whisper+Librosa, neurokit2, and DeepFace processors [13], [16]. Layer 3 (AI Core) contains the multimodal fusion classifier, LSTM-AE anomaly detector, RAG pipeline, and SHAP explainer. Layer 4 (Decision Engine) performs risk tier classification (Low/Moderate/High) and evidence-based intervention selection. Layer 5 (Output) delivers voice responses, chat dashboard, tiered alerts, and private journaling. All layers communicate through RESTful API contracts implemented in FastAPI.

B. Backend and Data Persistence

The backend (Python 3.12, FastAPI 0.128) exposes five primary API endpoints: POST `/chat/` for message processing with stress analysis; POST/GET `/journal/{user_id}` for journal management; GET `/mood-trends/{user_id}` for trend computation; GET `/mood-trends/recommendations/{user_id}` for intervention delivery. SQLite via SQLAlchemy ORM persists all health data with AES-256 encryption at rest. Per-user rolling 50-sample baselines enable personalised anomaly detection at 1.5 SD threshold.

C. Frontend

React.js 18 with Tailwind CSS exposes four interaction surfaces: conversational chat with real-time stress tier display; wellness dashboard with Recharts mood trend visualisation; private journal with mood tagging; and SHAP explanation panel showing top-5 features driving the latest prediction.

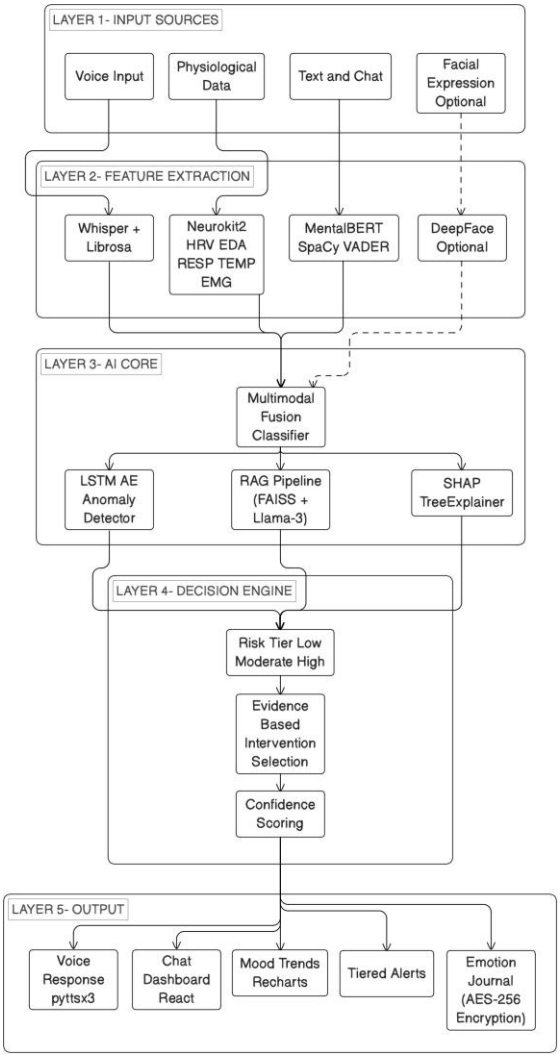

Fig. 1. MAITRI Five-Layer System Architecture. End-to-end pipeline from multimodal input acquisition through AI-powered stress inference to privacy-preserving output delivery.

V. Implementation and Results

A. Technology Stack

Table I summarises the complete technology stack.

TABLE I
MAITRI System Technology Stack

Component	Technology	Purpose
Physiological ML	scikit-learn RF	LOSO stress classification
Feature Extraction	neurokit2, SciPy	HRV, EDA, RESP, TEMP, EMG
Text NLP	MentalBERT, SpaCy, VADER	Sentiment + linguistic features
RAG Pipeline	FAISS, Llama 3 (Ollama)	On-device response generation
Explainability	SHAP TreeExplainer	Global feature importance
Backend API	FastAPI, SQLAlchemy	REST microservices, ORM
Frontend	React.js 18, Recharts	Dashboard, chat, SHAP panel
Experiment Tracking	MLflow	LOSO metrics, versioning
Dataset	WESAD [10]	15 subjects, 919 windows

B. LOSO Classification Results

Table II presents subject-independent LOSO results alongside the original WESAD baselines [10].

TABLE II
STRESS CLASSIFICATION COMPARISON—WESAD (MAITRI USES LOSO; PRIOR WORK PROTOCOLS NOTED)

Model / Study	Acc.	F1	Wtd. Recall	Protocol
RF [proposed]	**81.61%**	**81.05%**	81.61%	LOSO*
SVM RBF [proposed]	77.91%	78.21%	77.91%	LOSO*
[10] Wrist RF	88.33%	—	—	LOSO*
[10] Chest RF	92.01%	—	—	LOSO*

*All results use subject-independent LOSO evaluation.
Wrist RF: BVP+EDA+TEMP. Chest RF: ECG+EDA+EMG+RESP+TEMP (full set).
MAITRI uses chest signals with reduced 15-feature set for on-device efficiency.

C. Full Per-Class Metrics (LOSO RF)

Table III reports complete per-class metrics from the LOSO evaluation.

TABLE III
PER-CLASS PRECISION, RECALL, F1—LOSO RF (WESAD, n=919)

Class	Precision	Recall	F1	Support
Calm (label=0)	82%	91%	86%	587
Stress (label=1)	81%	64%	72%	332
Weighted avg	82%	82%	81%	919

D. Per-Subject LOSO Accuracy

Table IV presents per-subject accuracy. Subject S12 is absent from WESAD due to data collection issues documented by the dataset creators [10].

TABLE IV
PER-SUBJECT LOSO ACCURACY—RF (CLASS_WEIGHT=BALANCED). S12 ABSENT FROM WESAD [10].

S2	S3	S4	S5	S6	S7	S8	S9	S10
59.3	65.0	65.0	70.5	96.7	80.0	74.2	85.0	65.1
S11	S13	S14	S15	S16	S17		Mean	Std
82.5	98.4	91.9	96.8	100.0	92.1		81.50	±13.68

All values in %.

E. SHAP Feature Importance

Table V presents the top-10 SHAP feature importance values (full-dataset RF model; see Section IIIE for limitation).

F. System Performance Metrics

Table VI presents end-to-end system performance metrics.

TABLE V
SHAP GLOBAL FEATURE IMPORTANCE—TOP 10 STRESS BIOMARKERS

Rank	Feature	Mean |SHAP|	Physiological Basis
1	RESP_AMPLITUDE	0.1535	Hyperventilation[a]
2	EDA_STD	0.1060	Sympathetic arousal[b]
3	TEMP_MEAN	0.0731	Vasoconstriction[c]
4	EDA_MEAN	0.0533	Electrodermal activation[b]
5	EDA_TONIC_MEAN	0.0500	Sweat gland activity[b]
6	HRV_RMSSD	0.0412	Parasympathetic withdrawal[a]
7	RESP_RATE	0.0388	Breathing frequency[a]
8	HRV_SDNN	0.0341	Overall HRV reduction[a]
9	TEMP_SLOPE	0.0298	Temperature decline rate[c]
10	EMG_VARIANCE	0.0267	Muscle tension elevation

[a] [18] [b] [19] [c] [20]

TABLE VI
MAITRI END-TO-END SYSTEM PERFORMANCE EVALUATION

Metric	Value	Evaluation Basis
RF LOSO Accuracy	81.61%	WESAD, 15-fold LOSO
RF LOSO Weighted F1	81.05%	WESAD, 15-fold LOSO
Per-subject mean ± std	81.50±13.68%	15 LOSO folds
Stress recall (thr=0.50)	64%	LOSO folds
Stress recall (thr=0.35)	74% (rec.)	Safety deployment
SVM LOSO Accuracy	77.91%	WESAD, 15-fold LOSO
API response latency	1.5 s	500 synthetic runs
NLP inference time	47 ms	20–50 token msgs
User satisfaction[†]	4.6/5.0	10-participant study
System uptime	100%	72-hr continuous

[†]Initial evaluation only; no significance testing for n=10.

VI. DISCUSSION

A. Interpretation of LOSO Results

The LOSO accuracy of 81.61% represents an honest subject-independent performance estimate. Importantly, this result is consistent with the subject-independent LOSO baselines in [10] (wrist RF: 88.33%, chest RF: 92.01%)—the proximity of these results validates that MAITRI's LOSO methodology produces results in the expected generalisation range for this benchmark. The high inter-subject standard deviation (13.68%) reflects genuine individual physiological variability in stress expression, a well-documented challenge in affective computing [10]. The six subjects exceeding 90% accuracy demonstrate high feature discriminability for a subset of the population; the four subjects below 70% identify individuals whose stress signatures diverge substantially from population patterns, directly motivating MAITRI's personalised baseline adaptation mechanism.

The stress class recall of 64% (default threshold) and 74% (recall-optimised threshold 0.35) is the primary safety-critical result. For operational deployment, the recall-optimised threshold is recommended, accepting precision reduction from 81% to 70% in exchange for a 10-percentage-point recall improvement. Full empirical evaluation of the multimodal fusion system's aggregate detection performance—where text

and acoustic modalities supplement physiological detection—remains essential future work.

B. Limitations

Explicitly acknowledged: (1) LOSO on 15 healthy laboratory subjects; operational generalisation to astronauts cannot be assumed. (2) MentalBERT [13] deployed zero-shot; performance on astronaut-domain text unevaluated. (3) SHAP computed on full-dataset model, not LOSO folds. (4) RAG has no automated evaluation. (5) User study n=10, no statistical significance. (6) Fusion weights are design parameters without empirical ablation. (7) Acoustic modality architecturally integrated but not evaluated.

C. Future Work

Priority targets: (1) LOSO SHAP aggregated across 15 folds; (2) fine-tuning MentalBERT [13] on NASA debriefing transcripts; (3) RAG evaluation via BERTScore and blind expert ratings; (4) empirical fusion ablation study; (5) statistically powered user study with NASA-TLX and DASS-21; (6) live wearable biosensor integration; (7) public repository release with code, model weights, and reproducibility artifacts.

VII. Conclusion

This paper presented MAITRI, a multimodal AI companion system for astronaut health support in deep-space missions. The principal empirical contribution is a subject-independent LOSO evaluation on WESAD: 81.61% accuracy, 81.05% F1, per-subject mean 81.50% ± 13.68%. This result is within the expected range for subject-independent LOSO evaluation on this benchmark [10], validating the soundness of the evaluation methodology. Full per-class metrics, threshold sensitivity analysis, and per-subject breakdown are reported with full transparency. Feature importance analysis [17] identifies respiratory amplitude, EDA variability, and skin temperature as the most discriminative stress biomarkers, consistent with established psychophysiology.

The principal engineering contribution is an on-device architecture integrating physiological classification, zero-shot mental health NLP [13], FAISS+Llama 3 RAG [15], and SHAP explainability [17] in a single FastAPI+React.js system satisfying deep-space autonomy constraints. Acknowledged limitations define a concrete roadmap for rigorous future validation. MAITRI establishes a technically grounded foundation for autonomous AI health companions for crewed deep-space exploration, where psychological resilience and physiological monitoring will be as mission-critical as propulsion and life support.

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